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Academic Year		2025/2026	**Semester**	Spring
Department		Computer Engineering		
Exam	**Type**	Quizz	**Duration**	
	Date	2.06.2026	**Time**	13:00
Course	**Name**	Remote Sensing Technologies	**Code**	BMI3242
	Instructor	Asst. Prof. Dr. Alpay DORUK		
Student	**Name-Surname**		**Signature**	
	Number			

LO /PO	**1/1**								
Question (Score)	**1 (100)**								
Received Score									
Grade	**Written**		**Numeric**						
Exam Rules*									

Write a Python program that classifies land cover in a multispectral image using the k-means clustering algorithm. The program should process a given multispectral image and output a classified image where each pixel is assigned to the nearest cluster center.

Requirements:

1. Use the `sklearn.cluster.KMeans` class to perform k-means clustering.
2. Assume the multispectral image has four bands: Red, Green, Blue, and Near-Infrared (NIR).
3. The number of clusters (k) should be set to 5 for this assignment.
4. Use the Euclidean distance to measure the similarity between pixels and cluster centers.
5. As the output you should show the Red Band, Green Band, Blue Band, NIR Bands, RGB Combined, K-Means Clustering with Euclidian Distance, Correlation Matrix between bands and NDVI images.

Dataset: You can use any publicly available multispectral image dataset (for example, the Landsat 8 or Sentinel-2 datasets) or the image file in the Homework document for this assignment.